Quotation reference: **SQT/19-20/1891R** **Date: 07th Aug 2019**

Garrison Engineer (I) R&D East
C V Raman Nagar
DRDO Complex
Bangalore

Kind Attn : Mr. N Bhoyar

Contact No: +91 9823841275
Email ID : nbhoyar50@gmail.com

Project: **CABS DRDO, Old Airport, Bangalore - Karnataka**

Subject : Supply & Installation of NIHVA Make Sliding Hangar Door

Dear Sir,

Thank you for giving us an opportunity to participate with our offer on the above project. We enclose herewith our proposal which includes:

A. Requirement
B. Technical Details
C. Terms and Conditions
D. Pricing

We wish you success with your project and hope that we may be part of your current & future projects as a solution provider. We will follow up this quotation with a call to you within the next few days.

If you have any questions after reading this proposal, or if you need more information, please do not hesitate to contact us.

Kind Regards,

Ms. Varsha
NIHVA Technologies Pvt. Ltd
187/G Block, 1st Floor, Thermax Chowk,
Above Renault showroom, Chinchwad,
PUNE - 411019
Mob: +91 8551091333

A. Requirement Hangar Door:

Clear opening Width (M)	Clear Opening Height (M)	Qty in No
Total :33	Max :13	01

1. **Size of each door Clear Opening – 33M (W) x 13 M (H)**
2. **Quantity of Doors – 01 No`s**
3. **Type of Door – Motorised Multi-leaf**
4. **No of Leaves – 12 No`s (Bi parting Door with each side 6 No of Leaves)**

B. Technical Details:

1. DOOR CONSTRUCTION

The door leaf frames shall be designed to withstand as per required wind speed / wind load. The frames shall be fabricated out of heavy-duty mild steel sections / profiles. The frames shall be of modular design and all sections shall be pre-fabricated at works, ready for assembly at site.

2. PROTECTIVE COATING

After fabrication and before transportation all mild steel members shall be cleaned and one coat of primer applied. Before final assembly & installation at site. All bolts, nuts, washers shall be galvanised. Primer and Paint is synthetic enamel

3. SHEET CLADDING

1 mm Galvalume sheet.

4. VISION PANEL

One vision Panel at alternate panel shall be provided.

5. DOOR MOVEMENT

Each door leaf frame shall be supported at the bottom on three (3) load bearing wheels. The wheels are made of steel specially machined to run on bottom rail track. The wheels are fitted with double row sealed ball bearings. All wheels are bolted to the door frame and can be easily removed for maintenance. All the door leaves are guided at the top with three (3) heavy-duty steel ball bearing rollers.

6. BOTTOM TRACKS

The bottom tracks shall comprise of required runs of steel rails mounted on transverse support and alignment bearers along with fixing bolts etc. The bottom track assembly shall be assembled and aligned at site before grouting. The bottom tracks shall extend up to the extremities of the door-parking bay on one sides of the hangar building.

7. TOP GUIDE TRACKS

The top guides shall comprise of required sets of guide ways mounted on transverse support to the hangar structure. The door support shall be on brackets and dependent on the hangar building. The top guide tracks shall extend up to the extremities of the door-parking bay on both sides. Parking bay is excluded in the quote.

8. BUFFERS

Heavy-duty buffers shall be provided at the parking bay extremities of each bottom track. These buffers shall act as emergency mechanical stoppers in case of the doors overshooting their limits.

9. DRIVE ARRANGEMENT

A 3-phase 415 Volts, AC, 50 Hz electric motor of suitable rating shall drive each master leaf.We shall use helical bevel gear motor which has better efficiency. The drive transmission between the motor and the drive wheel shall be through an efficient gearbox. Each leaf shall be master leaf.

10. ELECTRICAL OPERATION

The electrical operation of each door section shall be carried out fr om a control panel mounted on the inside of each leaf. The operator shall have to keep the OPEN or CLOSE press button control pressed to enable the door to operate. At each extremity the door section shall automatically STOP, as the control power shall be cut-off through proximity switches. An EMERGENCY STOP BUTTON (STAY-PUT TYPE) shall also be provided on
the control panel.

11. MANUAL OPERATION

In case of power failure the doors can be operated manually by disengaging the gearbox and manually operating the gear drive with the handle provided separately. The sequence of manual operation shall be same as the electric operations.

12. ELECTRICAL CONTROL SYSTEM

- **GENERAL**

The electrical control system shall be able to control all aspects of the door movements. The control system for each door section shall consist of a control panel, proximity switches, trailing / bunching cables etc.

- **ELECTRICAL POWER SUPPLY**

The incoming power supply to the door control panel shall be 415V, 3-phase, 50 Hz, 4 – wire & earth. Power supply to each master leaf shall be by means of a looped trailing / bunching cable system, mounted in the top guide track area from a junction box placed at the top extremities inside the hangar.

- **ELECTRICAL OPERATION**

The electrical operation of each door section shall be carried out from a control panel mounted inside each master leaf. The operator shall have to keep the OPEN / CLOSE push button pressed to enable the door to operate. At each extremity the door shall automatically STOP, as the control power shall be cut-off through proximity switches. An EMERGENCY STOP BUTTON (STAY-PUT TYPE) shall also be provided on the control panel.

- **CONTROL PANEL & SYSTEM**

Each door control panel shall consist of a VFD system mounted on panel, control fuse holders and fuses, indication lamps, interlock contactors, proximity switches, warning lights/hooters, bus bar system on the door after the main in-coming power junction box etc.

- **DOOR OPERATION SAFETY**

The following features shall be incorporated in the control system of the doors:

a) The control panel shall be located for easy operation and enable the operator to have a full view of the area during operation of the door.

b) The OPEN & CLOSE joy-sticks are constant pressure (HOLD to RUN) and the EMERGENCY STOP push button is STAY PUT type and has to be twisted to release.

c) Emergency proximity switch shall be provided at the extreme limits of travel just before the mechanical buffers. These emergency switches shall totally cut-off the control / power supply to the system in case the doors over shoot their normal limits.

d) Audio / Visual Alarm – Each door control panel shall be provided with a Flashing Light cum Sounder that shall start & continue to operate during the time the door is in operation.

e) Key Operation – An operating key switch shall be provided on each control panel to prevent un-authorised operation of the doors.

C. PRICING –

Sr no	Description	Price/Each in Rs	Qty No	Total Price in INR
Hanger Door: Size – (W)33 Mtr x (H)13 Mtr				
1	Supply of above Hanger Door	3,15,00,000	1	**3,15,00,000**
2	Installation Price	25,00,000	1	**25,00,000**
3	Door Mounting Structure (design, mfg, supply)	35,00,000	1	**35,00,000**
4	Lifting crane and Equipment's	25,00,000	1	**25,00,000**
5	**Civil Work – Client Scope**	25,00,000	1	**25,00,000**
			Grand Total	**4,25,00,000**

General Pre-Requisites Required at Site

Prerequisites:

a) Finished Flooring after curing (Without Water)
b) Electrical Supply on the site single phase/three phase points at different places.
c) Provide electrical cables near to the installation machinery.

Limit of Supply

a) Provide place to store the material to be installed.
b) Provide change & rest room for the workers.

Terms & Conditions:

Payment : 80% along with confirmed purchase order & balance before dispatch.

P & F Charges : @3% of total supply value

Transport : Extra as applicable

GST : Extra 18%

Validity : 15 days from the date of offer

Delivery : 18-20 weeks from the date of receipt of confirmed order, drawing approval.

Insurance : Extra as applicable

Warranty :12 months from date of Supply.

General conditions of Aircraft Hangar Door

1. Continuous uninterrupted access to any single door section or as agreed separately.

2. 400 volt / 50Hz 3 phase isolator to be fitted to within 1 mtr of the door main control panel. If not available at the start of installation, a temporary supply is acceptable. Any return visits to commission the doors due to permanent power not being available will be charged as appropriate.

3. Provision of 230 Volt power supply (by others) for hand tools.

4. A dry & secure area, not in direct sunlight, is to be provided for storage of the door components.

5. Access to site between 7 am to 6 pm six days per week. If this is not possible the installation programme may require amending.

6. The project time indicated does not take into account adverse weather conditions that may prevent safe and efficient working practices. We reserve the right to extend the period accordingly.

7. Any passes required for the project, are the responsibility of the Customer must be available, before commencement of the installation. Any delays caused by the non-availability of passes, will be charged accordingly.

Inclusions

1. A pre-installation inspection will be carried out, by our Project Manager, before our installers travel to site, so as to verify that the structural steelwork and electrical installation are as per our requirements. Our Project Manager will also check that the installation equipment and facilities are ready for installers.

2. Civil Work Required for the Door Rail mounting.

3. Lifting Required for Installation

4. Door Mouting Strcture required for Door

Exclusions

The following items are not within our scope of works and are to be provided by others at no cost to NIHVA.

1. A compliance statement, with supporting documentation, which states that the building deflections are in accordance with the criteria, as agreed between our design department and the steelwork contractor (to be done by main contractor).

2. Any passes required for the project, are the responsibility of the Customer and must be available, before commencement of the installation. Any delays caused by the non-availability of passes, will be charged accordingly.

3. Power supply and isolators adjacent to the control panels (to be done by electrical contractor).

4. A 400/230 power supply (permanent or temporary) must be provided (FOC) by client during the installation of the door for power tools and testing of the door sections during erection.

5. A safe storage / pre-assembly area for the door materials

6. Any additional tests or certification, which are not within our scope of works but may be required by the Consultant or Client, these will be carried out or provided, subject to the actual requirements but will be at additional cost.

Please feel free to Contact Mr. Arjun Patil on 08551096333 For any further Clarification /Information in this Concern.